

H4 — ML + Embedded Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: the team sees where every weight lives on the chip.

Steps: 1. Read the src/ runtime sources. 2. Find where each of these lives: activations, norm weights, the core, the output head, the PLE table. 3. Map them to the tiers: SRAM, PSRAM, flash. 4. Write the 1-page memory-layout map.

Deliverable: the memory-layout map as a markdown file in the club repo.

Verify: you can answer: why does the head sit in PSRAM and the table in flash?

Learning this week

Next-token prediction. Explain in two sentences why the model reads 6 table rows per token.

Resources

- src/: <https://github.com/slvDev/esp32-ai/tree/main/src>
- RESULTS.md (memory numbers): <https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md>

Done when

- [] Memory-layout map exists in the club repo
- [] You can say what the project does in one sentence